HOTEL CLOUD MANAGEMENT SYSTEM

Software Features & Workflow Requirements for Development Team

Project Type: Custom cloud-based hotel management system for internal use by one hotel.

Purpose: Build one connected system for hotel operations, bookings, guests, rooms, payments, GST/non-GST invoicing, accounts, expenses, housekeeping, maintenance, and reports.

1. Core Development Principle

The software must be one connected system. Data should be entered once and automatically update all relevant modules. The receptionist or accountant must not manually enter the same transaction separately in Bookings, Payments, Accounts, and Reports.

Example workflow:

Booking → Payment → Accounts Updated → Invoice → Check-out → Reports Updated

2. Main Features Required

Dashboard

- Today's check-ins and check-outs
- Current occupancy and available rooms
- Today's bookings, revenue, payments received, and expenses
- Pending guest payments
- Cash, bank, UPI and card summary
- Quick actions for booking, check-in, payment and expense

Room Management

- Room types and room numbers
- Standard room rates
- Room status: Available, Reserved, Occupied, Dirty, Cleaning, Inspection, Maintenance, Blocked
- Prevent booking of blocked or maintenance rooms
- Room status history

Booking Management

- Direct, walk-in, phone and WhatsApp bookings
- Booking calendar
- Room availability checking
- Guest selection/creation
- Check-in and check-out dates
- Room rate, discounts, extra bed and additional charges
- Advance and partial payments
- Booking statuses: Enquiry, Hold, Confirmed, Checked-in, Checked-out, Cancelled, No-show
- Double-booking prevention

Guest Management

- Guest profile with name, mobile, email and address
- ID details and document attachment
- Nationality and GSTIN when applicable
- Guest stay history
- Previous bookings and outstanding balances
- Duplicate guest detection using mobile/email where possible

Check-in and Check-out

- Guest and ID verification
- Room allocation and room change
- Additional guests and extra beds
- Additional charges during stay
- Final bill calculation
- Outstanding balance warning
- Room automatically changes to Dirty after checkout

Payments

- Cash, UPI, bank transfer, credit card, debit card and configurable methods
- Advance, partial and final payments
- Multiple payments against one booking/invoice
- Payment receipt and transaction reference
- Refund workflow
- Automatic update of booking balance, accounts and reports

GST and Non-GST Invoicing

- User selects GST Invoice or Non-GST Invoice
- Separate invoice numbering/prefix configuration
- GSTIN and customer GSTIN fields
- Configurable tax rates
- CGST + SGST or IGST as applicable
- Taxable value and tax breakdown
- PDF and print invoice
- Controlled cancellation/correction with audit trail
- Credit note/refund support where required

Accounts

- Automatic income from hotel operations
- Expense recording
- Cash, bank, UPI and card accounts
- Accounts receivable/outstanding balances
- Payment and refund tracking
- Income vs expenses
- Profit & Loss

Expenses

- Configurable categories such as salary, electricity, food purchase, maintenance and supplies
- Supplier/payee
- Payment method
- Bill/receipt attachment
- Approval status if required
- Automatic update to accounts and reports

Housekeeping

- Automatic task creation after checkout
- Room statuses: Dirty → Cleaning → Inspection (optional) → Available
- Staff assignment
- Priority and notes
- Completion tracking

Maintenance

- Maintenance issue reporting
- Priority, assigned person and status
- Maintenance cost tracking
- Open → In Progress → Completed workflow
- Room unavailable while under maintenance if required

Reports

- Daily/monthly bookings
- Occupancy
- Revenue
- Payment method summary
- Expenses
- Outstanding balances
- Profit & Loss
- GST summary
- Booking source
- Cancellations and refunds
- Room-wise and room-type performance
- ADR and RevPAR

Users, Permissions and Security

- Owner/Super Admin, Manager, Receptionist, Accountant and Housekeeping roles
- Role-based permissions
- Backend permission enforcement
- Audit/activity logs
- No silent deletion of financial records
- Secure login and password handling
- HTTPS and cloud backups
- Restore capability

3. Main End-to-End Workflow

1. Create or select Guest
2. Check room availability
3. Select room, dates and rate
4. Create booking
5. Receive advance payment if applicable
6. Booking status becomes Confirmed
7. Check-in guest and room status becomes Occupied
8. Add extra services/charges during stay
9. System updates guest folio and outstanding balance
10. Start check-out and calculate final bill
11. Receive final payment or record outstanding due
12. Select GST or Non-GST invoice
13. Generate invoice
14. Automatically update accounts and reports
15. Complete check-out
16. Room status becomes Dirty
17. Housekeeping completes cleaning
18. Room status becomes Available

4. Detailed Workflows

Booking Workflow

1. Search available rooms
2. Select room and dates
3. Create/select guest
4. Enter guest and stay details
5. Set rate and discounts
6. System calculates booking total
7. Record advance or partial payment if received

8. Confirm booking

9. Update booking calendar and room availability

Payment & Accounts Workflow

1. Receive payment

2. Select payment method

3. Enter amount and reference

4. Link payment to booking/invoice

5. Automatically update amount paid and balance due

6. Automatically update the correct Cash/UPI/Bank/Card account

7. Automatically update payment reports

GST / Non-GST Invoice Workflow

1. Final bill is ready

2. Select GST Invoice or Non-GST Invoice

3. For GST: calculate configured CGST/SGST or IGST as applicable

4. For Non-GST: do not add GST

5. Generate sequential invoice number

6. Create PDF/print invoice

7. Update invoice register, accounts and reports

Expense Workflow

1. Create expense

2. Select category

3. Enter supplier/payee and amount

4. Select payment method

5. Attach bill/receipt if available

6. Approve if approval workflow is enabled

7. Post expense

8. Automatically update accounts, cash/bank balance and Profit & Loss

Refund Workflow

1. Select original payment/invoice

2. Enter refund amount and reason

3. Require approval where configured

4. Record refund without deleting original payment

5. Update booking/invoice balance and financial accounts

6. Create audit log

Housekeeping Workflow

1. Guest checks out

2. Room becomes Dirty

3. Housekeeping task is created

4. Assign staff

5. Staff changes status to Cleaning

6. Cleaning completed

7. Optional inspection

8. Room becomes Available

Maintenance Workflow

1. Report issue

2. Set priority

3. Assign responsible person

4. Change status to In Progress

5. Block room if required

6. Record maintenance cost

7. Complete maintenance

8. Return room to Available after release

5. Critical Business Rules

- Enter each transaction once only. All connected modules must update automatically.
- Every payment must be traceable to its booking, guest, invoice, payment method, user and date/time.
- Do not permanently delete normal financial records. Use cancellation, void, reversal, refund or credit note processes with audit logs.
- Prevent double booking by validating room availability before confirmation.
- Room status changes must follow hotel workflows automatically.
- Historical bookings and financial records must remain traceable after changes.

- Use precise decimal/currency handling for all money calculations.
- Permissions must be enforced in the backend, not only hidden in the user interface.
- Invoice and tax configuration should be configurable rather than hard-coded.
- All important changes must record who made the change and when.

6. Database / Data Requirements

The development team should design connected data entities for:

- Users
- Roles
- Permissions
- Hotel Settings
- Room Types
- Rooms
- Room Status History
- Guests
- Guest Documents
- Bookings
- Booking Rooms
- Booking Status History
- Check-ins
- Check-outs
- Folio Charges
- Service Items
- Payments
- Payment Allocations
- Refunds
- Invoices
- Invoice Items
- Tax Details
- Credit Notes
- Accounts
- Account Transactions
- Expenses
- Expense Categories
- Housekeeping Tasks
- Maintenance Requests
- Attachments
- Audit Logs

7. Phase 1 – Must Have

- Login, users and permissions
- Dashboard
- Rooms and room statuses

- Guest management
- Booking calendar and booking management
- Walk-in and direct bookings
- Check-in/check-out
- Advance and partial payments
- Cash, UPI, bank and card payment recording
- GST and Non-GST invoicing
- PDF/print invoices
- Expenses
- Basic accounts and outstanding balances
- Housekeeping
- Maintenance basics
- Core booking, revenue, payment, expense, GST and Profit & Loss reports
- Audit logs and cloud backups

8. Phase 2 – Future Features

- OTA/channel manager integration
- Online booking engine
- WhatsApp, SMS and email automation
- Supplier and purchase management
- Inventory
- Restaurant/POS integration
- Advanced accounting integration
- Staff attendance and payroll
- Mobile application
- Dynamic pricing
- Advanced analytics and forecasting
- Multi-property support if needed later

9. Development Acceptance Tests

Test	Expected Workflow
Booking to Accounts	Create booking → Receive advance → Booking balance updates → Accounts update automatically → Report reflects payment.
GST Invoice	Create stay → Add charges → Generate GST invoice → Verify tax calculation → Verify accounts and GST report.
Non-GST Invoice	Create stay → Generate Non-GST invoice → Verify no GST is incorrectly added → Verify accounts and reports.
Checkout to Housekeeping	Check out guest → Room becomes Dirty → Cleaning task → Cleaning completed → Room becomes Available.
Expense to P&L	Record expense → Correct payment account changes → Expense report updates → Profit & Loss updates.
Refund	Record refund → Original payment remains traceable → Balance and accounts update → Audit log records the action.

10. Final Instruction to Development Team

Build this as one integrated cloud system, not as separate disconnected modules. Bookings, guests, payments, invoices, accounts, expenses, housekeeping and reports must use connected records and update automatically wherever applicable.

The final principle is: Enter data once → validate it → store it safely → update all relevant modules automatically → keep a complete audit trail.